

**YEAR 10 SCIENCE
BIOLOGICAL SCIENCES
REVISION**

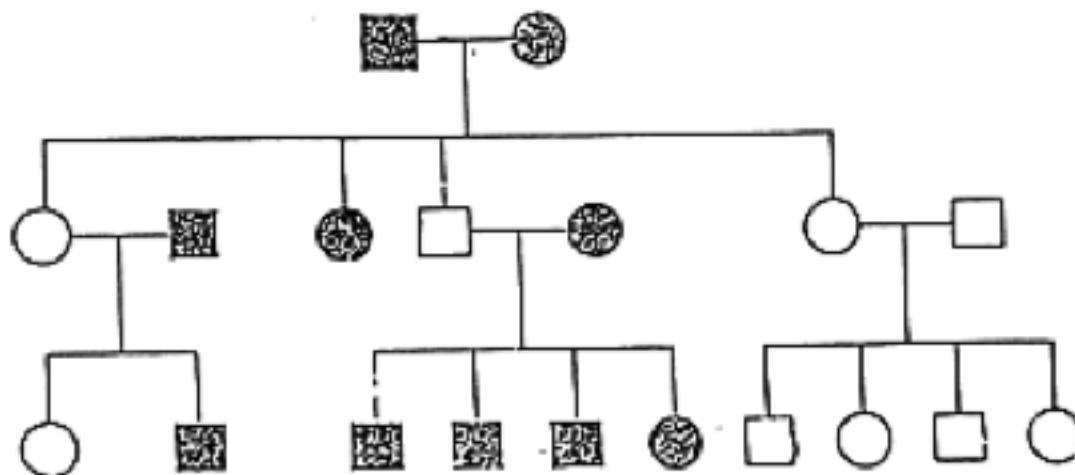
1.
 - (a) Give a brief description of the DNA molecule.
 - (b) Name the **4 bases** in a DNA molecule.
 - (c) Name the **base pairs** found in DNA.
 - (d) Draw the adjoining strand of DNA for the following.

- G - T - T - A - C - A - G - T
2.
 - (a) Which process of cell division is responsible for the growth and repair of cells?
 - (b) Which process is involved in producing the reproductive cells of an animal?
 - (c) What is the meaning of **haploid** and **diploid number**?
3. An organism has 4 chromosomes in its normal body cells. Draw a series of simple diagrams showing the stages involved in **mitosis**. *Name each stage* and state *what is happening* in each.

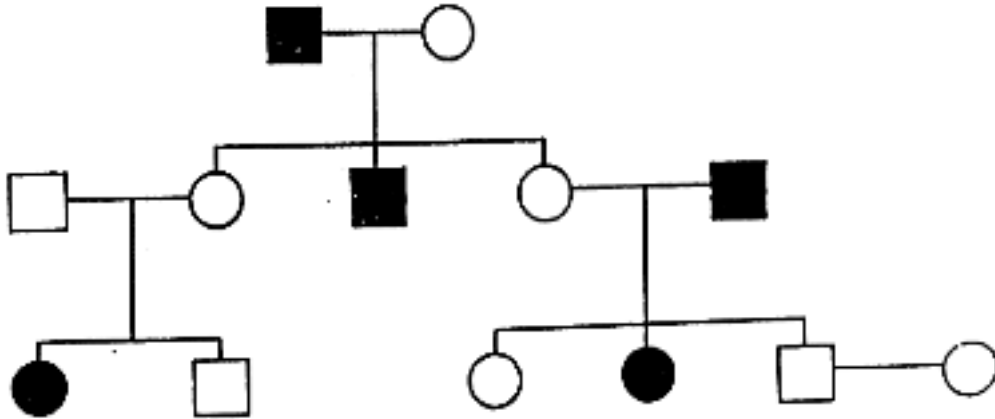
4. Meiosis occurs in reproductive organs in **two stages**. Draw a series of diagrams to show the stages involved. **Name each stage** and state **what is happening** in each.
5. (a) What is the difference between the male and female sex chromosome pair?
- (b) Who determines the sex of a child - the mother or father? Explain.
6. (a) Show, by punnett square, why half of any particular population would be expected to be female.
- (b) A family has had five girls born into it, and the mother is pregnant with the sixth child. What is the chance that the child will be a girl? Explain why.

7. Briefly describe the significance of the work of Gregor Mendel to our understanding of genetics.
8. Define the following terms.
- allele
 - pure-bred
 - hybrid
 - homozygous
 - heterozygous
 - genotype
 - phenotype
 - sex-linked
 - dominant
 - recessive
 - co-dominant
 - incomplete dominance
 - haploid
 - diploid.

9. Give one example of how **environment** can affect how a characteristic shows in an individual.
10. In humans, the ability to taste a chemical phenylthiocarbamide (PTU) is due to the presence of the dominant gene T.
- (a) Determine the genotype and phenotypes of the offspring produced by the marriage of a homozygous non-taster male and a heterozygous taster female.



11. The pedigree below shows the existence of straight hair in a family.



- (a) Is straight hair a dominant or recessive trait? Explain your answer.
- (b) Determine the genotype of all individuals.
12. Red-green colour blindness is a sex-linked, recessive disorder in humans. A colour-blind male marries a woman with normal colour vision. They have two girls, one of whom is colour blind, and two boys, one of whom is colour blind.

What is the most likely genotype of the mother? Set out an appropriate cross or pedigree to support your answer.

13. (a) Explain the difference between **co-dominance** and **incomplete dominance**. You may need to give an example of each to help your explanation.
- (b) In shorthorn cattle, red coat colour co-dominates with white coat colour, producing a roan colour in a heterozygous animal. Determine the F1 and F2 generation offspring for a red bull mated with a white cow. Set out your working.
- (c) In snapdragon flowers, red colour is incompletely dominant over white. Hybrid plants have pink flowers. Determine the genotypes and phenotypes for the offspring produced from a red flowered plant with a pink one.

14. In guinea pigs, rough coat dominates smooth coat. Determine the genotypes and phenotypes for the mating of two rough coated hybrid animals.
15. In humans, suppose that the gene that controls right-handedness is dominant over left-handedness. A heterozygous right-handed male and a left-handed female have four children, three of who are left-handed.
- (a) If R represents the gene for right-handed, use a Punnett square to determine the expected ratio of children that should be right-handed.
- (b) Suggest a reason for difference between the result in part (a) and the actual proportion of children in this family who are right-handed.

16. (a) Define the term **adaptation**.
- (b) There are three types of adaptations - give an example for each.
- (i) Behavioural
 - (ii) Structural
 - (iii) Physiological
17. Explain how each of the following examples are adaptations for survival.
- (a) Desert marsupials excrete small quantities of very concentrated urine.
 - (b) Polar bears have a thick fur coat.
 - (c) Frogs burrow into the ground during hot and dry weather.
18. Why is **genetic diversity** so important to the survival of a species?
19. What is the relationship between **mutation** and **genetic variation**?
20. What is meant by the term **natural selection**?
21. Myxomatosis virus was used to control rabbits in Australia but it is no longer effective. Explain why this is the case.

22. Describe how ***variation***, ***selection pressures*** and ***competition*** contribute to natural selection, and hence evolution.